

LOGIC AND PROOFS

Propositional Logic

Thuma E and Zimudzi E.

Department of Computer Science
University of Botswana

11 August, 2025

OUTLINE

1. PROPOSITIONAL LOGIC

Logic

Logic is the basis of all mathematical reasoning, and of all automated reasoning. It has practical applications to the design of computing machines, to the specification of systems, to artificial intelligence, to computer programming, to programming languages, and to other areas of computer science such as:

- design of computer circuits,
- the construction of computer programs,
- the verification of the correctness of programs,
- verify mathematical proofs using a computing system,
- etc.

Proposition

A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either true or false, but not both.

Examples of Propositions:

1. Gaborone is the capital city of Botswana.
2. Toronto is the capital of Zambia.
3. $1 + 1 = 2$.
4. $2 + 2 = 3$.

The area of logic that deals with propositions is called the **propositional calculus** or **propositional logic**. We use letters such as p , q , r , s , *etc* to denote propositional variables.

The **truth value** of a proposition is denoted by **T** if it is a true proposition, and by **F** if it is a false proposition. **Compound propositions** are formed from existing propositions using logical operators.

The following are not propositions

1. What time is it?
2. Read this carefully.
3. $x + 1 = 2$.
4. $x + y = z$

Constructing Compound Propositions

A compound proposition is comprised of propositions and one or more of the following connectives.

- | | | |
|------------------|-------------------|------------------|
| 1. NEGATION | $\neg$ | "NOT" |
| 2. CONJUNCTION | $\wedge$ | "AND" |
| 3. DISJUNCTION | $\vee$ | "OR" |
| 4. IMPLICATION | $\rightarrow$ | "IF, THEN" |
| 5. BICONDITIONAL | $\leftrightarrow$ | "IF AND ONLY IF" |

Definition: Negation

Let p be a proposition. The **negation** of p , denoted by $\neg p$ is the statement. "It is not the case that p ." The proposition is read "not p ."

The truth value of the negation of p , is the opposite of the truth value of p .

Truth Table for the Negation of a Proposition.	
p	$\neg p$
T	F
F	T

Figure: The Truth Table for the Negation of a Proposition

Definition: Conjunction

Let p and q be propositions. The **conjunction** of p and q , denoted by $p \wedge q$ is the proposition “ p and q .” The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

The Truth Table for the Conjunction of Two Propositions.		
p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

The Truth Table for the Disjunction of Two Propositions.		
p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Figure: The Truth Table for the Conjunction and a Disjunction

Definition: Disjunction

Let p and q be propositions. The **disjunction** of p and q , denoted by $p \vee q$ is the proposition “ p or q .” The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Definition: Exclusive or

Let p and q be propositions. The **Exclusive or** of p and q , denoted by $p \oplus q$ is the proposition that is true when exactly one of p and q is true and is false otherwise:

The Truth Table for the Exclusive Or of Two Propositions.		
p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

The Truth Table for the Conditional Statement $p \rightarrow q$.		
p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Figure: The Truth Table for the Exclusive Or and a Conditional

Definition: Implication(Conditional Statement)

Let p and q be propositions. The conditional statement (also called an implication) $p \rightarrow q$ is the proposition “if p , then q .” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise. In the conditional statement $p \rightarrow q$, p is called the hypothesis (or antecedent or premise) and q is called the conclusion (or consequence).

Other common ways to express $p \rightarrow q$:

“if p , then q ”

“if p , q ”

“ p is sufficient for q ”

“ p implies q ”

“ p only if q ”

“a sufficient condition for q is p ”

Converse, Contrapositive, and Inverse

From the conditional statement $p \rightarrow q$, we can construct 3 new conditional statements.

1. **Converse** ($q \rightarrow p$) The proposition $q \rightarrow p$ is called the converse of $p \rightarrow q$.
2. **Inverse** ($\neg p \rightarrow \neg q$) The proposition $\neg p \rightarrow \neg q$ is called the inverse of $p \rightarrow q$.
3. **Contrapositive** ($\neg q \rightarrow \neg p$) The contrapositive of $p \rightarrow q$ is the proposition $\neg q \rightarrow \neg p$. The contrapositive always has the same truth value as $p \rightarrow q$.

When two compound propositions always have the same truth values, regardless of the truth values of its propositional variables, we call them **equivalent**. Hence, **a conditional statement and its contrapositive are equivalent**.

Example: Find the contrapositive, the converse, and the inverse of the conditional statement

1. "The home team wins whenever it is raining."
2. "It is raining is a sufficient condition for me not going to Riverwalk."
3. "The lecturer is happy when you complete your tutorial."

Definition: Biconditional

Let p and q be propositions. The biconditional statement (aka bi-implications) $p \leftrightarrow q$ is the proposition “ p if and only if q .” The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise.

Other common ways to express $p \leftrightarrow q$:

“ p is necessary and sufficient for q ”

“if p then q , and conversely”

“ p iff q .” “ p exactly when q ”

The Truth Table for the Biconditional $p \leftrightarrow q$.		
p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Figure: The Truth Table for the Biconditional $p \leftrightarrow q$

Logic and Bit Operations

Computers represent information using bits. A bit (binary digit) is a symbol with two possible values, namely, 0 (zero) and 1 (one). A bit can be used to represent a truth value, because there are two truth values, namely, true and false. That is, 1 represents T (true), 0 represents F (false). A variable is called a Boolean variable if its value is either true or false. Consequently, a Boolean variable can be represented using a bit.

Table for the Bit Operators <i>OR</i>, <i>AND</i>, and <i>XOR</i>.				
x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Bit string

A bit string is a sequence of zero or more bits. The length of this string is the number of bits in the string.

Example: 1011 is a bit string of length 4.

Example: 10001011 is a bit string of length 8.

We use the symbols $\vee$, $\wedge$, and $\oplus$ to represent the bitwise OR, bitwise AND, and bitwise XOR operations, respectively.

01 1011 0110	
11 0001 1101	
11 1011 1111	bitwise <i>OR</i>
01 0001 0100	bitwise <i>AND</i>
10 1010 1011	bitwise <i>XOR</i>

Figure: Bitwise OR, bitwise AND, and bitwise XOR of the strings below.

The End